

# Wireless Communication Exam Notes

EX 715 (BEI) / EX 751 (BEX) · IV-I · TU, IOE · tiers over 22 papers, 2070–2082 BS

## 4. Modulation–Demodulation Methods in Mobile Communications

4 hrs · asked in **20 of 22 papers** · 5 syllabus sub-topics · **no paper has ever set a numerical here** (one worked example is given anyway, §4.4).

Read this first:

- **This is a drawing chapter.** Almost every question says +Fig, block diagram or constellation diagram. Marks come from the **diagram**, not the prose.
- Four blocks carry the chapter, in order of return:
  - **OFDM** **TOP 9/22** → transmitter + receiver block diagram, always 4 or 8 marks.
  - **Spread spectrum, DSSS and FHSS** **TOP 7/22** → definition + block diagram.
  - **MSK / GMSK** **TOP 6/22** → why GSM picked GMSK.
  - **QPSK** **HOT 4/22** → transmitter, receiver, constellation.
- Everything else (BPSK, DPSK, OQPSK,  $\pi/4$ -QPSK, QAM) is a **one-paper** question, but each was worth 6 to 8 marks when it came.
- **Learn 3 diagrams cold:** QPSK transmitter/receiver pair, OFDM transmitter/receiver pair, DSSS + FHSS transmitter pair. They answer roughly 30 marks of the paper.
- FHSS appears here as a *modulation* and again in Ch 7 as **FHMA**, a *multiple access* method. Same block diagram, different framing. Draw it once, label it for whichever chapter is asked.

### 4.1 Why modulate, why digital, and how schemes are judged

Only one PYQ lives in this section, and it is worth 2 marks. Everything else here is **vocabulary the rest of the chapter assumes**:  $\eta_p$ ,  $\eta_B$ , the trade-off between them, and what a constellation diagram actually plots. Skim it once, then never look at it again.

## Advantages of digital modulation over analog 2

79 Bh 2 · always a rider on the GMSK question, never asked alone

### Modulation itself

- Process of impressing message on high-freq carrier suitable for transmission.
- Translates baseband (modulating) signal  $\rightarrow$  bandpass (modulated) signal.
- **Why:** antenna size  $\propto \lambda$ , so baseband speech at 3 kHz needs km-long antennas.
- **Why:** shifting to different carriers lets many users share one medium (multiple access).
- **Why:** higher freq  $\rightarrow$  longer range, smaller hardware.

### Power efficiency $\eta_p$

- Ability to hold message quality at low power (low SNR).
- $\eta_p = E_b/N_0$  at a stated BER, usually  $10^{-5}$ .
- $E_b$  = energy per bit,  $N_0$  = noise power spectral density.
- Trade-off: BER  $\uparrow$  as  $E_b/N_0 \downarrow$ .
- **Lower  $E_b/N_0$  for the same BER = more power efficient.**

### Choosing a scheme for a mobile link

- Low BER at low SNR (handset battery is finite).
- Resistance to interference (ACI, CCI) and multipath fading.
- Minimum occupied bandwidth (spectrum is bought, not free).
- Cheap, small, low-power to implement in the handset.
- **No scheme meets all four**, so every standard is a trade-off.

### Constellation (signal space) diagram

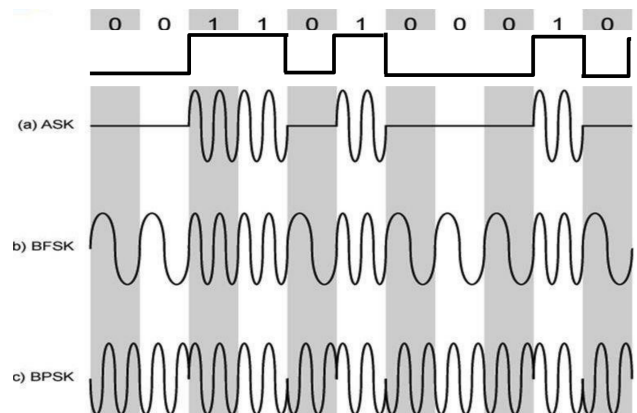
- Graphical plot of the complex envelope of every possible symbol.
- $x$  axis = in-phase component,  $y$  axis = quadrature component.
- Distance of a point from origin = amplitude, so distance<sup>2</sup> = symbol power.
- **Distance  $d$  between the two nearest points sets noise immunity.**
- $P_e \propto Q\left(\frac{d}{\sqrt{2N_0}}\right)$ , so small  $d \rightarrow$  high BER.
- Farthest point from origin = peak power the amplifier must deliver.

### Digital over analog 2

- Regenerable: noise does not accumulate, receiver rebuilds clean symbols.
- Resistant to fading, noise, interference.
- Voice + data + video multiplexed on one channel.
- Encryption possible  $\rightarrow$  security and privacy.
- Error control coding detects and corrects errors (Ch 6).
- Built in DSP software, not analog hardware  $\rightarrow$  cheap, small, reprogrammable.
- Equalization and diversity apply cleanly (Ch 5).
- **Cost:** needs more bandwidth per message and needs synchronisation.

### Bandwidth efficiency $\eta_B$

- Ability to carry data in a limited band.
- $\eta_B = R/B$  bps/Hz,  $R$  = data rate,  $B$  = RF bandwidth.
- Ceiling set by Shannon:  
 $\eta_{B(\max)} = C/B = \log_2(1 + S/N)$ .
- **The fundamental trade-off:** improve  $\eta_p$  and  $\eta_B$  gets worse, and the reverse. No scheme wins both.
- M-ary buys  $\eta_B$  by spending  $\eta_p$ .



The three binary keying waveforms against one data stream: (a) ASK, (b) BFSK, (c) BPSK. SST, Modulation extra note

## 4.2 Review: analog modulation and line coding syllabus, no PYQ yet

### Amplitude family

- **AM (DSB-FC):**  $B = 2f_m$ . Carrier carries no information, so at most 33 % of power is useful. Simple envelope detector.
- **DSB-SC:** carrier suppressed,  $B = 2f_m$ . All power in sidebands, but needs coherent detection.
- **SSB:** one sideband only,  $B = f_m$ . **Half the bandwidth, half the power**, but sharp filters and exact carrier recovery needed.
- **VSF:** one full sideband + vestige of the other,  $B \approx 1.25f_m$ . Compromise used for TV video where DC content forbids true SSB.

### Line coding: what it is for

- Maps bits onto baseband pulses before modulation.
- Requirements: no DC content, self-clocking, spectrally compact, error detection possible, transparent to any bit pattern.

### Angle family

- **FM:** instantaneous freq varies with message. Carson's rule  $B = 2(\Delta f + f_m)$ .
- Constant envelope  $\rightarrow$  non-linear class-C amplifier usable  $\rightarrow$  power efficient.
- **Capture effect:** stronger of two co-channel FM signals suppresses the weaker.
- Wideband FM trades bandwidth for SNR  $\rightarrow$  why **every 1G system used FM**.
- **PM:** phase varies with message. FM and PM are inter-convertible through an integrator or differentiator.

### The four to know

- **NRZ:** 1 = high for full bit, 0 = low. Compact, but has DC and no clock in long runs. **GSM uses bipolar NRZ ahead of the GMSK filter.**
- **RZ:** pulse returns to zero mid-bit. Self-clocking, costs twice the bandwidth.
- **Manchester:** transition mid-bit, 1 = hi $\rightarrow$ lo, 0 = lo $\rightarrow$ hi. No DC, always self-clocking, twice the bandwidth. Used in 10 Mbps Ethernet.
- **AMI (bipolar):** 0 = zero level, 1 = alternating  $\pm$ . No DC, and a run of same-polarity ones flags an error.

## 4.3 Linear modulation: BPSK, DPSK, QPSK, OQPSK, $\pi/4$ -QPSK

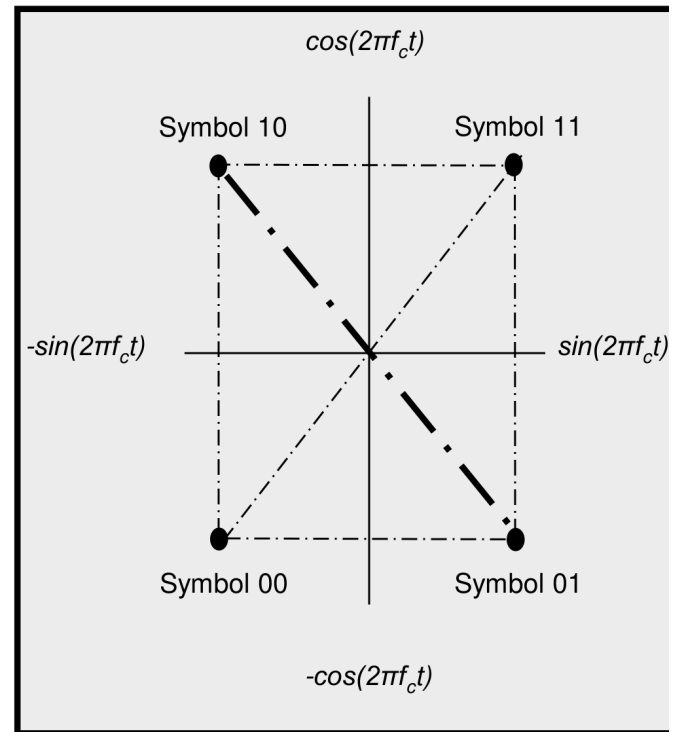
Linear means the transmitted amplitude follows the modulating signal linearly. Bandwidth efficient, but envelope varies  $\rightarrow$  **needs a linear (backed-off, inefficient) power amplifier**. That is exactly why mobile handsets moved to the constant-envelope family in §4.4.

## QPSK: transmission and detection process + Fig, constellation diagram 6 7 8

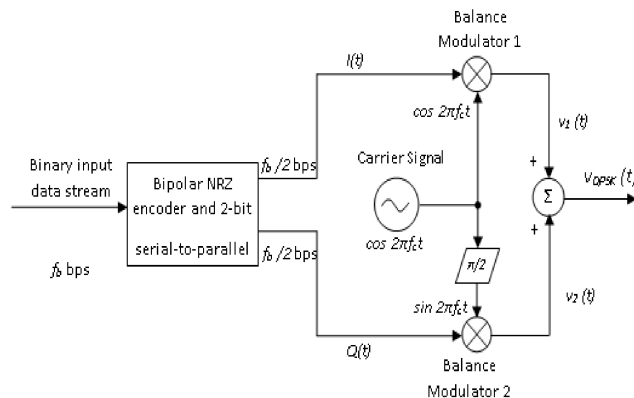
**HOT 4/22** 75 Bh 6, 72 Ma 8, 81 Bh 8, 73 Bh 7 · answer = two block diagrams + one constellation + six lines

### Idea

- M-ary PSK with  $M = 4$ , so **2 bits per symbol** (a dibit).
- Carrier phase takes one of 4 equally spaced values  $\pi/4, 3\pi/4, 5\pi/4, 7\pi/4$ .
- $s_i(t) = \sqrt{\frac{2E_s}{T_s}} \cos(2\pi f_c t + (2i - 1)\frac{\pi}{4})$ ,  $i = 1, 2, 3, 4$ .
- Symbol period  $T_s = 2T_b \rightarrow$  **half the bandwidth of BPSK for the same bit rate**.
- Null-to-null RF bandwidth  $= R_b$  (BPSK needs  $2R_b$ ), so  $\eta_B = 1$  bps/Hz.
- **BER identical to BPSK**: the dibit splits into two independent BPSK streams on orthogonal carriers,  $P_e = Q\left(\sqrt{2E_b/N_0}\right)$ .
- So QPSK gets double the bandwidth efficiency **for free**. That is the whole selling point.
- Symbols are **Gray coded** (00, 01, 11, 10 around the circle) so a wrong symbol costs only 1 bit.



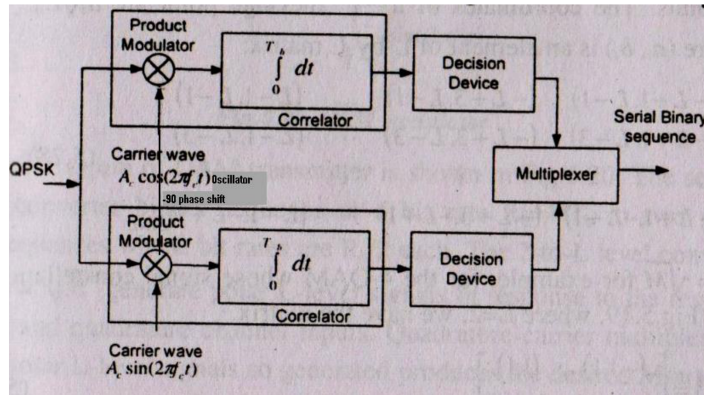
QPSK constellation: 4 points, one per dibit, spaced  $90^\circ$ . Distance to origin  $= \sqrt{E_s}$ , distance between neighbours  $= \sqrt{2E_s} = 2\sqrt{E_b}$ . SST, Modulation extra note



QPSK transmitter. SST, Fig 3.2.3

### Transmission process 4

1. Binary stream at  $f_b$  bps enters a **bipolar NRZ encoder**  $\rightarrow \pm 1$  levels.
2. **Serial-to-parallel converter** splits it: even-numbered bits  $\rightarrow I(t)$ , odd-numbered bits  $\rightarrow Q(t)$ , each at  $f_b/2$  bps.
3.  $I(t)$  multiplied by  $\cos 2\pi f_c t$  in balance modulator 1.
4.  $Q(t)$  multiplied by  $\sin 2\pi f_c t$  (carrier through a  $90^\circ$  shifter) in balance modulator 2.
5. **Summer** adds the two  $\rightarrow$  QPSK signal, one of 4 phases per dibit.
6. Output BPF limits the spectrum to the allocated channel.



QPSK receiver, a pair of coherent correlators. SST, Modulation extra note

**Detection process 4**

1. Received QPSK splits into two arms (coherent detection, so carrier recovery circuit supplies cos and sin references).
2. Upper arm: product modulator with  $\cos 2\pi f_c t$ , then **integrate-and-dump** over 0 to  $T_b \rightarrow$  output  $x_1$ .
3. Lower arm: product modulator with  $\sin 2\pi f_c t$ , then integrator  $\rightarrow$  output  $x_2$ .
4. **Decision device**, threshold 0 V:  $x_1 > 0 \rightarrow 1$ ,  $x_1 < 0 \rightarrow 0$  on the I channel, same rule for  $x_2$  on the Q channel.
5. **Multiplexer** interleaves the two recovered streams  $\rightarrow$  original binary sequence.

**Say "coherent" out loud.** QPSK cannot be envelope detected, the receiver needs the carrier phase.

**The weakness that produced OQPSK and  $\pi/4$ -QPSK**

- When *both* bits of a dibit change, the phase jumps  $180^\circ$  and the **envelope passes through zero**.
- After the transmitter's band-limiting filter this zero crossing becomes a deep amplitude dip.
- A non-linear (efficient) amplifier hard-limits that dip  $\rightarrow$  the filtered sidelobes **grow back**  $\rightarrow$  adjacent channel interference returns.
- Fixing this without giving up bandwidth efficiency is the whole story of §4.3 and §4.4.

**BPSK against QPSK 5**

73 Ma 5

| Parameter          | BPSK                        | QPSK                                   |
|--------------------|-----------------------------|----------------------------------------|
| Bits per symbol    | 1                           | 2 (dibit)                              |
| Symbol period      | $T_s = T_b$                 | $T_s = 2T_b$                           |
| Constellation      | 2 points, $180^\circ$ apart | 4 points, $90^\circ$ apart             |
| Null-to-null RF BW | $2R_b$                      | $R_b$                                  |
| $\eta_B$           | 0.5 bps/Hz                  | <b>1 bps/Hz</b>                        |
| Bit error rate     | $Q(\sqrt{2E_b/N_0})$        | $Q(\sqrt{2E_b/N_0})$ , <b>the same</b> |
| Phase jumps        | $180^\circ$ only            | $0^\circ, \pm 90^\circ$ or $180^\circ$ |
| Circuit            | one balance modulator       | two, plus S/P and $90^\circ$ shifter   |

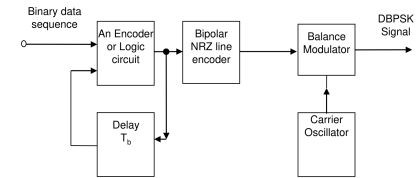
**The one-line answer:** QPSK carries twice the data in the same band at the same BER and the same  $E_b/N_0$ , paying only in circuit complexity.

## DPSK modulation: transmitter and receiver; PN sequence and why it is used 4+2+2

PIN 1/22 72 Ma 4+2+2 · the PN half of this question is really \$4.7 material, cross-refer it

### What DPSK is

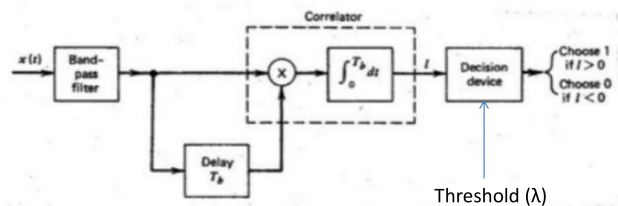
- **Non-coherent form of PSK**: no synchronous carrier needed at the receiver.
- Information sits in the **difference between the phases of two successive symbols**, not in the absolute phase.
- Encoding rule: symbol **1** = no phase transition, symbol **0** =  $180^\circ$  transition.
- So the present bit is decoded by comparing against the previous bit.
- $P_e = \frac{1}{2}e^{-E_b/N_0}$ , about **3 dB worse** than coherent BPSK at high SNR.



DPSK transmitter: encoder +  $T_b$  delay ahead of an ordinary BPSK modulator. SST, Modulation extra note

### Transmitter 2

1. Binary sequence  $m_k$  enters a **logic (differential encoder)** circuit.
2. A **1-bit delay**  $T_b$  feeds the previous encoded bit  $d_{k-1}$  back to it.
3. Encoder output  $d_k$  toggles when  $m_k = 0$  and holds when  $m_k = 1$ .
4. **Bipolar NRZ encoder**  $\rightarrow \pm 1$  levels.
5. **Balance modulator** multiplies by the carrier  $\rightarrow$  DPSK out.



DPSK receiver: the signal is its own phase reference. SST, Modulation extra note

### Receiver 2

1. **Bandpass filter** centred on  $f_c$  removes out-of-band noise.
2. Signal is multiplied by **its own  $T_b$ -delayed copy** in the correlator.
3. **Integrate and dump** over one bit gives an output proportional to  $\cos \phi$ , where  $\phi$  is the phase difference between the two.
4.  $\phi = 0 \rightarrow$  output positive  $\rightarrow$  decide **1**.  
 $\phi = \pi \rightarrow$  output negative  $\rightarrow$  decide **0**.
5. **Decision device** at threshold  $\lambda = 0$  V rebuilds the sequence.

**No carrier recovery circuit anywhere in this diagram.** That is the mark the examiner is looking for.

|                                    |    |       |   |   |       |   |   |       |   |   |
|------------------------------------|----|-------|---|---|-------|---|---|-------|---|---|
| Binary data<br>b(t)                | 0  | 0     | 1 | 0 | 0     | 1 | 0 | 0     | 1 | 1 |
| Diff. encoded<br>data d(k)         | 1* | 0     | 1 | 1 | 0     | 1 | 1 | 0     | 1 | 1 |
| Phase of DPSK                      | 0  | $\pi$ | 0 | 0 | $\pi$ | 0 | 0 | $\pi$ | 0 | 0 |
| Shift Diff. encoded<br>data d(k-1) | 1  | 0     | 1 | 1 | 0     | 1 | 1 | 0     | 1 | 1 |
| Phase of shift DPSK                | 0  | $\pi$ | 0 | 0 | $\pi$ | 0 | 0 | $\pi$ | 0 | 0 |
| Phase com. o/p                     | -  | -     | + | - | -     | + | - | -     | + | + |
| Detected Binary seq                | 0  | 0     | 1 | 0 | 0     | 1 | 0 | 0     | 1 | 1 |

Differential encoding worked through: the starred bit is the arbitrary reference. Row 2 is  $d_k$ , row 3 its phase, last row the recovered data. SST, Modulation extra note

### DPSK against BPSK

- **Bandwidth**  $f_b$  against  $2f_b \rightarrow$  DPSK halves it.
- **Carrier at demodulator:** not required against required.
- **Error probability:** higher against low.
- **Bit detection:** over two successive bit intervals against a single interval.
- **Circuit:** simpler receiver, but the encoder adds transmitter complexity.

### (+) and (-)

- (+) No carrier recovery  $\rightarrow$  cheap, small, fast to acquire.
- (+) Works where the channel rotates phase faster than a PLL can track.
- (-)  $P_e$  roughly 3 dB worse than coherent BPSK.
- (-) **Errors propagate in pairs:** one bad symbol corrupts the reference for the next.

### PN sequence and why it is used 2+2 (full treatment in §4.7)

- **PN (pseudo-noise) sequence** = a coded sequence of 1s and 0s that looks random to anyone without the generator, but is **completely deterministic and repeatable**.
- Generated by an  $m$ -stage shift register with XOR (parity) feedback; maximal-length period =  $2^m - 1$  chips.
- **Properties:** 1s and 0s roughly equiprobable; sharp autocorrelation peak and near-zero elsewhere; low cross-correlation between different codes; adding a shifted copy gives the same sequence in a new phase; easy to generate and synchronise.
- **Why used:** spreads the spectrum, so the signal hides under the noise floor (low probability of intercept) and shrugs off narrowband jamming.
- **Why used:** nearly orthogonal codes let many users share one band at once  $\rightarrow$  CDMA.
- **Why used:** gives the receiver a unique pattern to correlate against, which is both the addressing scheme and the synchronisation reference.

## OQPSK transmitter and receiver; why $\pi/4$ -QPSK is preferred over OQPSK 5+2

PIN 1/22 71 Bh 5+2

### Offset QPSK (OQPSK), also called staggered QPSK

- Same circuit as QPSK, with one change: the  $Q$  arm is delayed by  $T_b = T_s/2$  before its balance modulator.
- Effect: the even and odd streams can no longer change at the same instant.
- So only one of  $I$  or  $Q$  switches at a time  $\rightarrow$  maximum phase change is  $\pm 90^\circ$ , never  $180^\circ$ .
- The envelope therefore never passes through zero, and band-limiting produces only shallow dips instead of nulls.
- Hard-limiting or class-C amplification no longer regenerates the sidelobes, so OQPSK keeps its spectral occupancy in a real transmitter.
- Phase transitions happen every  $T_b$  instead of every  $T_s$ , i.e. twice as often but half as violent.
- PSD, BER and bandwidth are identical to QPSK. The gain is purely in envelope behaviour.

### Transmitter 2.5

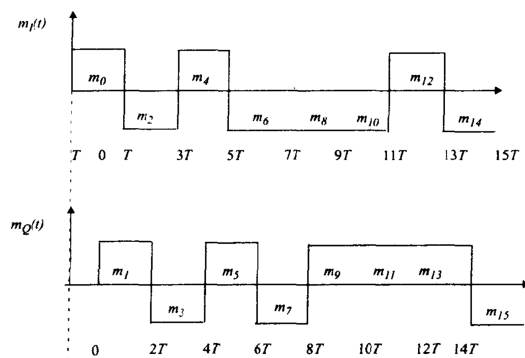
1. NRZ  $\rightarrow$  serial-to-parallel into  $m_I(t)$  and  $m_Q(t)$ .
2. Insert a  $T_b$  delay in the  $Q$  branch.
3.  $m_I \times \cos 2\pi f_c t$ , delayed  $m_Q \times \sin 2\pi f_c t$ .
4. Sum, then BPF  $\rightarrow$  OQPSK out.

### Receiver 2.5

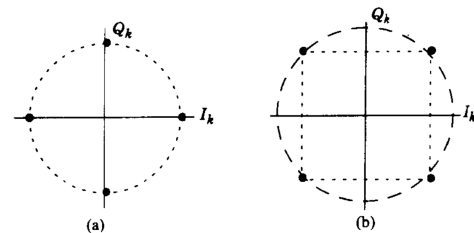
1. Coherent, exactly the QPSK receiver: BPF, carrier recovery, two correlators, two decision circuits.
2. The  $T_b$  stagger is removed by delaying the  $I$  decision (or advancing  $Q$ ) before the multiplexer, so the dibits re-align.
3. Symbol timing recovery is taken from the transitions, which now occur every  $T_b$ .

### $\pi/4$ -QPSK, the compromise

- Two QPSK constellations, offset from each other by  $\pi/4$ , used on alternate symbols.
- Signalling points therefore take 8 possible phase states, but only 4 are available in any one symbol.
- Maximum phase change is limited to  $\pm 135^\circ$ : worse than OQPSK's  $90^\circ$ , much better than QPSK's  $180^\circ$ .
- Every symbol carries a phase transition (no  $0^\circ$  option), which makes clock recovery easy.
- Usually sent differentially encoded as  $\pi/4$ -DQPSK.
- Used in IS-136 (US digital cellular), PDC and PH5 (Japan), TETRA.



OQPSK time offset:  $m_I(t)$  changes at even multiples of  $T$ ,  $m_Q(t)$  at odd multiples. Note the half-symbol offset. Rappaport Fig 5.30



$\pi/4$ -QPSK: constellation (a) and constellation (b) are used on alternate symbols, giving the 8-state pattern below with all allowed transitions drawn. Rappaport Fig 5.31

### Why $\pi/4$ -QPSK is preferred over OQPSK 2

1. It can be detected non-coherently. Because it is differentially encoded, a differential or discriminator detector works. OQPSK must be detected coherently, and the half-symbol offset makes differential detection impossible.
2. Non-coherent detection  $\rightarrow$  no carrier recovery circuit  $\rightarrow$  far simpler, cheaper, faster-acquiring receiver.
3. Better in multipath spread and fading, where a coherent receiver cannot hold a phase reference. This is the mobile channel exactly.
4. Guaranteed transition every symbol  $\rightarrow$  reliable symbol timing recovery even on long runs of identical dibits.
5. Cost accepted: envelope variation is larger than OQPSK's, so a mildly linear amplifier is still needed.

verified/added

#### 4.4 Constant envelope modulation: BFSK, MSK, GMSK

**Constant envelope** (also called non-linear) means carrier amplitude never changes, whatever the data. **Why mobile radio loves it:** a class-C amplifier can be run into saturation at 60 to 70 % efficiency instead of a linear amplifier backed off to 20 %, which is the difference between a day and an afternoon of battery life. Also: very little out-of-band spillage, simpler receiver, and high immunity to random FM noise and Rayleigh fading.

##### BFSK, the starting point

- Two carrier frequencies  $f_{c1}$  (mark) and  $f_{c2}$  (space) carry 1 and 0.
- $s_i(t) = \sqrt{2E_b/T_b} \cos(2\pi f_{ci}t)$ ,  $i = 1, 2$ .
- Constant amplitude  $\rightarrow$  constant envelope.
- Detected **coherently** (two correlators, compare outputs) or **non-coherently** (two BPFs + envelope detectors + comparator).
- (+) simple, insensitive to amplitude fluctuation, low  $P_e$ .
- (-) **needs more bandwidth than ASK or PSK**.
- (-) If the two oscillators are free-running the phase jumps at every transition  $\rightarrow$  spectral widening, large sidelobes.
- Fix: **CPFSK**, one oscillator whose frequency is shifted, so phase stays continuous.

##### MSK = the best CPFSK

- Definition 1 (as CPFSK):** a special case of CPFSK in which the peak frequency deviation is exactly **one quarter of the bit rate**,  $\Delta f = R_b/4$ , giving modulation index  $h = 2\Delta f T_b = 0.5$ .
- $h = 0.5$  is the **smallest index for which the two tones are orthogonal**,  $f_1 - f_2 = 1/(2T_b)$ . Hence “**minimum shift**”.
- Orthogonal  $\rightarrow$  uncorrelated  $\rightarrow$  the receiver separates them cleanly.
- Definition 2 (as OQPSK):** MSK is OQPSK in which the rectangular data pulses are replaced by **half-cycle sinusoidal** pulses.
- That pulse shaping is what makes the phase change *linear* and bounded to  $\pm\pi/2$  over a bit interval, so phase is continuous and smooth.
- Also called **Fast FSK**: the frequency shift is only half that of ordinary FSK.
- Good envelope, good spectral efficiency, good BER, no bandpass filter needed.

#### MSK as a special type of CPFSK and of OQPSK; compare OQPSK, MSK and GMSK; which suits GSM

2 4 3+3+2

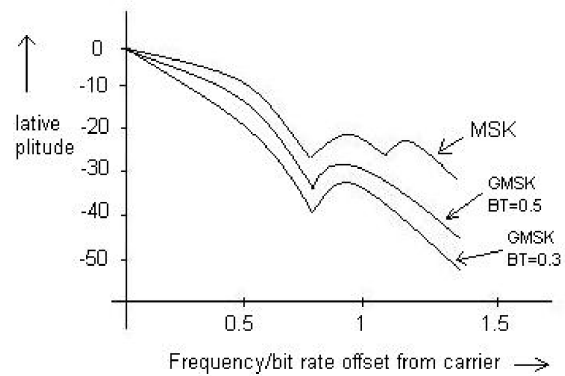
**HOT 4/22** 80 Ba, 81 Ba 3+3+2, 80 Ch 2, 76 Bh 4 · MSK and GMSK together are asked in 6 of 22 papers

|                   | OQPSK                  | MSK                | GMSK                          |
|-------------------|------------------------|--------------------|-------------------------------|
| Family            | linear                 | constant envelope  | constant envelope             |
| Pulse shape       | rectangular            | half-cycle sine    | Gaussian                      |
| Null-to-null BW   | $R_b$                  | $1.5R_b$           | below $R_b$                   |
| 99 % power BW     | narrowest of the three | $1.20R_b$          | $0.86R_b$ at $BT = 0.25$      |
| Sidelobe roll-off | $f^{-2}$ , slowest     | $f^{-4}$           | fastest of the three          |
| Envelope          | near-constant          | <b>constant</b>    | <b>constant</b>               |
| Power efficiency  | $Q(\sqrt{2E_b/N_0})$   | the same           | slight loss from ISI          |
| Amplifier needed  | mildly linear          | class C, saturated | class C, saturated            |
| Detection         | coherent only          | coherent or not    | <b>FM discriminator works</b> |

**Read the table as a story:** OQPSK has the narrowest main lobe but the slowest-decaying tails, MSK trades a wider main lobe for tails that die four times faster, GMSK then squeezes the main lobe back down with a Gaussian filter and pays for it in ISI.

### Which is suitable for GSM, and why 2

- **GMSK with  $BT = 0.3$ .** Three reasons, in order of weight:
  1. **Constant envelope** → the handset can use a saturated class-C power amplifier → battery life and cheap RF.
  2. **Compact spectrum** → fits the 200 kHz GSM carrier with acceptable adjacent channel interference. MSK alone leaks too much.
  3. **Detectable non-coherently** by a simple FM discriminator, so the receiver stays cheap.
- $BT = 0.3$  is the chosen compromise: tight enough spectrum, ISI penalty still small enough to be equalized out.



Occupied RF bandwidth

| BT       | 90%  | 99%  | 99.9% | 99.99% |
|----------|------|------|-------|--------|
| 0.2 GMSK | 0.52 | 0.79 | 0.99  | 1.22   |
| 0.25GMSK | 0.57 | 0.86 | 1.09  | 1.37   |
| 0.5 GMSK | 0.69 | 1.04 | 1.33  | 2.08   |
| MSK      | 0.78 | 1.20 | 2.76  | 6.00   |

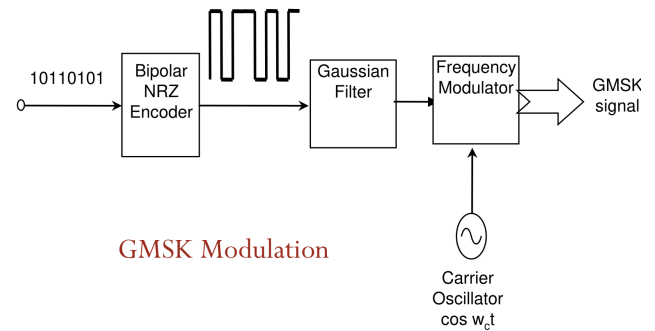
PSD of MSK against GMSK at three  $BT$  values, plus the occupied RF bandwidth table in units of  $R_b$ . **MSK's tails do not fall fast enough** to protect the adjacent channel, which is the entire reason GMSK exists. SST Ch 4

## GMSK transmission and detection process, block diagram and constellation

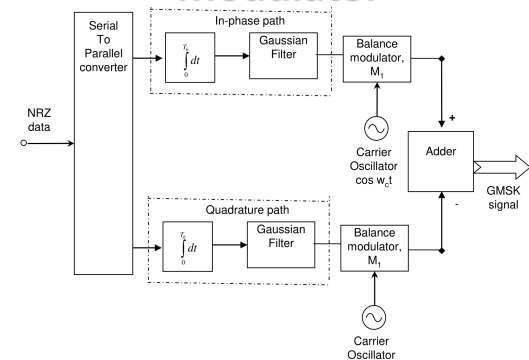
**HOT 4/22** 79 Bh **2+6**, 81 Ba **8**, 76 Bh **4**, 70 Ma **8** · 79 Bh pairs it with “advantages of digital modulation”, see §4.1

### What GMSK is

- MSK preceded by a **premodulation Gaussian low-pass filter**.
- The Gaussian filter has **no zero crossings and no overshoot**, so it smooths the rectangular data into a shape that suppresses the high-frequency content responsible for MSK’s sidelobes.
- Transfer function  $H(f) = \exp\left[-\frac{\ln 2}{2} \left(\frac{f}{B}\right)^2\right]$ , where  $B$  is the **3 dB bandwidth** of the filter.
- The filter is fully described by  $B$  and the symbol duration  $T$ , so GMSK is named by its  **$BT$  product**.  $BT = \infty$  is plain MSK.
- **Smaller  $BT$** : tighter spectrum, sidelobes fall away faster, but the pulse spreads over more symbols  $\rightarrow$  more **ISI**  $\rightarrow$  higher BER.
- GSM:  $BT = 0.3$ . CDPD and Mobitex also use GMSK.
- Constellation: same 4 points as QPSK, but the trajectory between them **follows the unit circle** instead of cutting across it, which is the constant envelope drawn.



GMSK transmitter, FM route. SST Ch 4



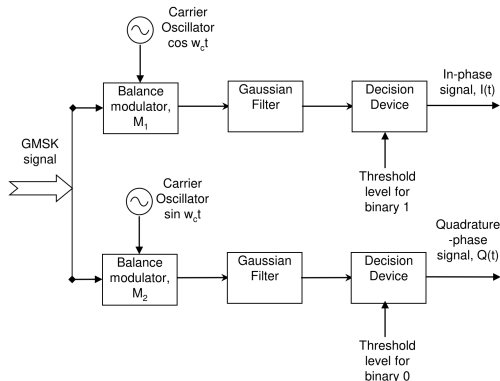
GMSK transmitter, I-Q route. SST, Fig 3.2.17

### Transmission process 3

1. Data  $\rightarrow$  **bipolar NRZ encoder**  $\rightarrow \pm 1$  rectangular pulses.
2.  $\rightarrow$  **Gaussian LPF** of bandwidth  $B$ , which rounds every pulse edge.
3.  $\rightarrow$  **FM modulator** set to modulation index exactly 0.5, driven by the carrier oscillator  $\cos \omega_c t$ .
4. Output is GMSK: constant amplitude, continuous smoothly varying phase.
5. **I-Q alternative** (right-hand figure above): the filtered signal is integrated, split into in-phase and quadrature paths, each Gaussian-filtered and multiplied by  $\cos \omega_c t / \sin \omega_c t$ , then subtracted. **Advantage: the index stays at exactly 0.5 with no tuning or drift.**

### Detection process 3

1. **Coherent (orthogonal) detector**, drawn left: GMSK splits into two arms.
2. Each arm has a **balance modulator** fed by  $\cos \omega_c t$  or  $\sin \omega_c t$ , then a **Gaussian filter** matched to the transmitter's.
3. Each arm's **decision device** compares against its own threshold, giving  $I(t)$  and  $Q(t)$ , which are recombined into the bit stream.
4. **Non-coherent alternative**: a simple **FM discriminator** followed by an integrate-and-dump. Much cheaper, small BER penalty. This is what a GSM handset actually does.



GMSK detector, orthogonal coherent form. SST, Fig 3.2.18

**Worked example: sizing the Gaussian filter** verified/added (tutorial problem, never a PYQ)**PROBLEM** SST tutorial · not from any paper

Find the 3-dB bandwidth for a Gaussian low pass filter used to produce 0.25 GMSK with a channel data rate of  $R_b = 270$  kbps. What is the 90 % power bandwidth in the RF channel?

**Step 1: symbol duration**

- Binary GMSK, so one bit per symbol and  
 $T = 1/R_b = 1/270,000 = 3.7 \mu\text{s}$ .

**Step 2: 3-dB bandwidth from the  $BT$  product**

- “0.25 GMSK” means  $BT = 0.25$ , so  
 $B = 0.25/T = 0.25/3.7 \mu\text{s} = 67.57 \text{ kHz}$ .

**Step 3: 90 % power RF bandwidth from the table**

- Read the occupied RF bandwidth table (§4.4 figure): row  $BT = 0.25$ , column 90 % gives  $0.57R_b$ .
- RF BW =  $0.57 \times 270,000 = 153.9 \text{ kHz}$ .

**Answer**

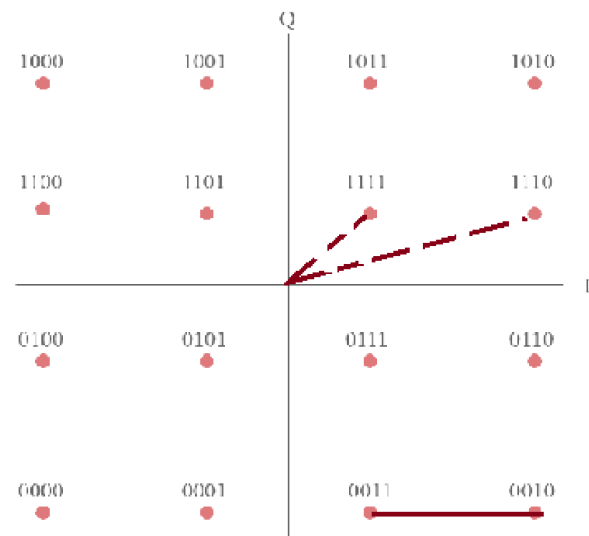
- 3-dB filter bandwidth  $B = 67.57 \text{ kHz}$ .
- 90 % power RF bandwidth = **153.9 kHz**.

**Notes on this problem**

- Real GSM uses  $BT = 0.3$  and  $R_b = 270.833 \text{ kbps}$** , not the rounded values here. Answer with whatever the paper prints.
- The table is in units of  $R_b$ , so always multiply the tabulated figure by the bit rate. Forgetting that is the standard slip.
- Sanity check: 153.9 kHz sits inside GSM’s 200 kHz carrier spacing, which is exactly the design intent.

**4.5 M-ary modulation: MPSK, MFSK and QAM****The M-ary idea**

- Send one of  $M = 2^n$  distinct waveforms per signalling interval, carrying  $n$  bits.
- Symbol duration  $T_s = nT_b$ , so the symbol rate drops by  $n$  for the same bit rate.
- Bandwidth needed falls by a factor of  $n$** , i.e.  $\eta_B$  rises to  $n$  bps/Hz in principle.
- Cost: the  $M$  points crowd together,  $d_{\min}$  shrinks, so **more power is needed** for the same BER.  $\eta_p$  falls.
- MPSK**: all  $M$  points on one circle of radius  $\sqrt{E_s}$ , differing only in phase. Constant envelope, but  $d_{\min} = 2\sqrt{E_s} \sin(\pi/M)$  collapses as  $M$  grows.
- MFSK**:  $M$  orthogonal frequencies. Here bandwidth *grows* with  $M$  while power efficiency *improves*: the opposite trade to MPSK.
- QAM**: vary amplitude *and* phase, so points fill a 2-D grid instead of a circle.
- 3G and later lean on these: EV-DO adapts between QPSK, 8-PSK and 16-QAM by channel quality.



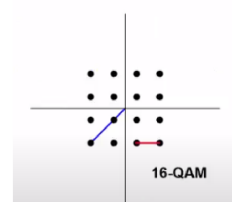
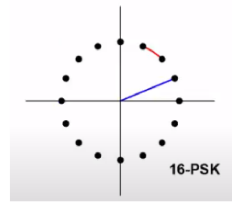
**16-QAM**, a  $4 \times 4$  grid, 4 bits per symbol. Nearest-neighbour distance is the same everywhere, which is why the grid beats the circle. SST Ch 4

## Why do we need M-ary QAM? Compare 16-QAM and 16-PSK using constellation diagrams 2+6

PIN 1/22 77 Ch 2+6

### Why M-ary QAM is needed 2

- Spectrum is licensed and finite, data demand is not. Binary schemes cap out near 1 bps/Hz.
- M-ary raises  $\eta_B$  to  $\log_2 M$  bps/Hz without extra bandwidth.
- But **M-ary PSK alone fails at high  $M$** : all points sit on one circle, so  $d_{\min}$  falls like  $\sin(\pi/M)$  and the power cost explodes.
- **QAM uses the radial dimension too**. Spreading points over a 2-D grid gives a much larger  $d_{\min}$  for the same average power.
- So M-ary QAM is the only practical way to reach 4, 6 or 8 bits per symbol on a band-limited channel.



16-PSK (left): 16 points on one circle, all at the same power, spacing  $22.5^\circ$ .

16-QAM (right): 16 points on a  $4 \times 4$  grid, three different amplitudes. Same 4 bits per symbol either way. SST Ch 4

| Parameter                           | 16-PSK                                                                                                           | 16-QAM                                |
|-------------------------------------|------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| Bits per symbol                     | 4                                                                                                                | 4                                     |
| Bandwidth                           | same                                                                                                             | same (identical symbol rate)          |
| Constellation                       | 16 points, one circle                                                                                            | $4 \times 4$ grid, 3 amplitude levels |
| Envelope                            | <b>constant</b>                                                                                                  | varies                                |
| Amplifier                           | non-linear, efficient                                                                                            | <b>linear, backed off</b>             |
| $d_{\min}$ at equal peak power      | smaller                                                                                                          | <b>larger</b>                         |
| Power for the same $d_{\min}$       | <b>1.6 dB more peak power</b>                                                                                    | reference                             |
| Power at equal $d_{\min}$ (average) | <b>4.19 dB more average power</b> <span style="background-color: #90EE90; padding: 0 2px;">verified/added</span> | reference                             |
| Noise immunity                      | equal, if $d_{\min}$ is equalised                                                                                | equal, at less power                  |

### The number the examiner wants

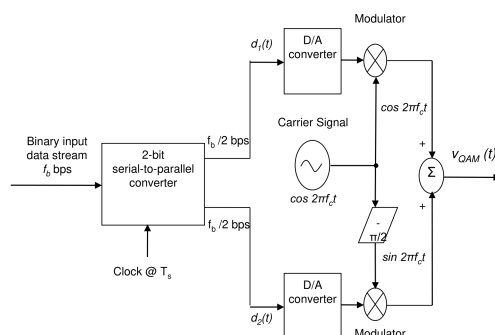
- **At the same minimum Euclidean distance (the same noise immunity), 16-QAM needs 1.6 dB less peak power than 16-PSK.**
- Measured on *average* power instead, the gap is **4.19 dB** verified/added.
- Both figures describe the same fact: the grid packs 16 points more economically than the circle.
- **Careful:** the lecture note's line "noise immunity is same" is only true when the two are compared at equal  $d_{\min}$ . At equal transmit power 16-QAM is clearly better.

### So why is 16-PSK used at all

- Constant envelope  $\rightarrow$  the transmitter can saturate its amplifier.
- 16-QAM's amplitude variation forces a linear amplifier with 6 to 8 dB of back-off, which **gives back much of the 4.19 dB**.
- Rule of thumb: QAM on the downlink (base station has mains power), constant envelope on the uplink (handset does not).
- Both need coherent detection and accurate gain control.

### QAM modulator, walked through

1. Binary stream at  $f_b$  enters a **2-bit serial-to-parallel converter** (4-bit for 16-QAM), clocked at the symbol rate  $T_s$ .
2. Each half feeds a **D/A converter** that turns the bit group into a multi-level signal  $d_1(t)$  and  $d_2(t)$ .
3.  $d_1(t)$  multiplies  $\cos 2\pi f_c t$ ,  $d_2(t)$  multiplies  $\sin 2\pi f_c t$  (carrier through a  $\pi/2$  shifter).
4. Summer adds the two  $\rightarrow$  a signal whose amplitude and phase both carry data.
5. **Demodulator** is the QPSK receiver with A/D converters instead of 1-bit decision devices, plus a carrier recovery circuit (raise to 4th power, BPF at  $4f_c$ , divide by 4).



QAM modulator. SST, Fig 3.2.21

## 4.6 OFDM the single most examined topic in this chapter

Orthogonal frequency division multiplexing. One high-rate carrier replaced by many low-rate ones, spaced so they overlap without interfering. It is the air interface of WiFi, WiMAX, LTE and 5G, and it is asked in **9 of the 22 papers** here, always with a block diagram attached.

### OFDM operation, block diagram of transmitter and receiver 4 8

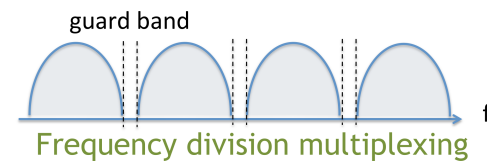
**TOP 9/22** 74 Bh 8, 80 Bh 8, 78 Ch 8, 71 Ma 8, 70 Bh 4, 79 Ch 4 (Tx only), 80 Ch 4 (Rx only), 82 Bh 4+4, 70 Ma 8 · **9 of 22 papers.**  
**Nothing else in the chapter comes close.**

#### The problem OFDM solves

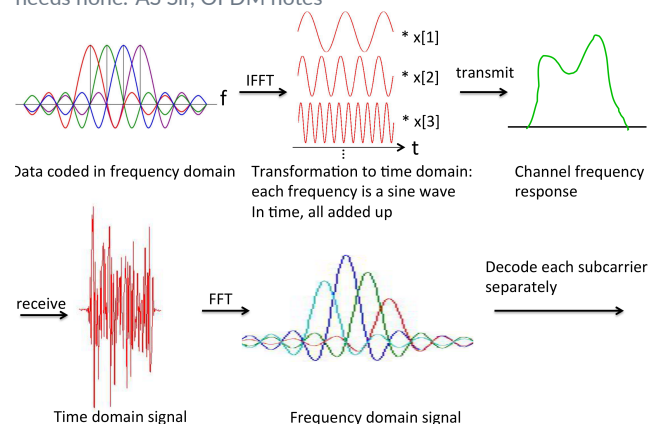
- Push the bit rate up on a **single carrier** and the symbol period  $T_s$  shrinks.
- Once  $T_s$  approaches the channel's **delay spread**  $\sigma_\tau$ , echoes of symbol  $n$  land on top of symbol  $n+1$  → severe **ISI** and frequency-selective fading.
- Fixing that in the time domain needs an equalizer with a great many taps, which is expensive and must adapt continuously.
- **OFDM's answer: do not make the symbol shorter, make it longer, and send many of them side by side.**

#### The principle 4

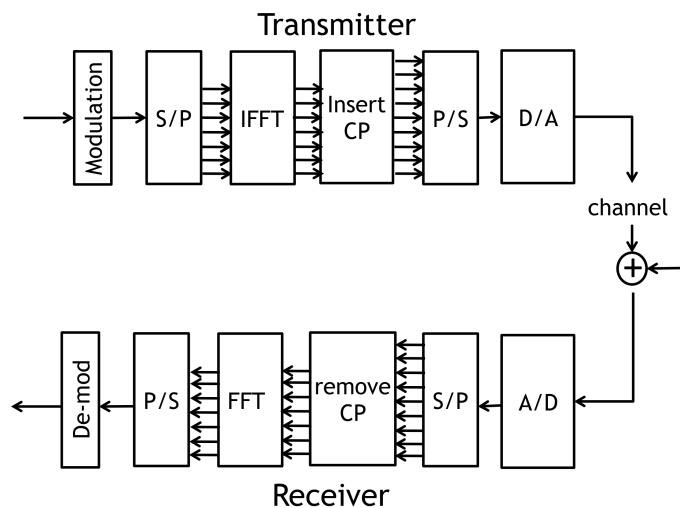
- Split the high-rate stream into  $N$  **parallel low-rate substreams**.
- Each substream modulates its own **subcarrier**, using QPSK or QAM.
- Each subcarrier's symbol period becomes  $NT_s$ , now **much longer than the delay spread** → each subchannel sees **flat fading**, not selective.
- Flat fading → one complex multiply per subcarrier undoes the channel. **The many-tap equalizer disappears.**
- Subcarrier spacing is set to  $\Delta f = 1/T$ , which makes them **orthogonal**: each one's spectrum has a null exactly where every other one peaks.
- So the spectra may **overlap without interfering**, and no guard bands are needed. Against FDM that is roughly a **50 % bandwidth saving**.
- The whole bank of  $N$  modulators is computed as one **IFFT**, and the bank of demodulators as one **FFT**. No oscillator bank, no filter bank.



**The figure that answers "OFDM principle".** FDM (top) wastes spectrum on guard bands. OFDM (bottom) overlaps orthogonal subcarriers and needs none. AS Sir, OFDM notes



Data is coded in the **frequency** domain, IFFT turns it into a time waveform, the channel acts, FFT recovers each subcarrier separately. AS Sir, OFDM notes



Draw exactly this for the 8-mark question. Transmitter on top, receiver below, channel and noise between them. AS Sir, OFDM notes

### Cyclic prefix, the detail that earns marks

- Even with long symbols, the tail of one OFDM symbol still smears into the next.
- A blank guard interval would work, but the hardware must transmit continuously and the exact delay spread is unknown.
- Solution: copy the last  $T_g$  of the symbol and glue it on the front.**
- Two effects:
  - Any echo arriving within  $T_g$  falls inside the copied portion, so it corrupts only the prefix, which is thrown away  $\rightarrow$  **ISI removed**.
  - Linear convolution with the channel becomes *circular* convolution, so after the FFT each subcarrier is simply scaled:  $Y[k] = H[k]X[k] \rightarrow$  **one-tap equalization**.
- Requirement:  $T_g >$  maximum delay spread. In 802.11, CP: data = 1 : 4.
- Side benefit: the packet still decodes even if detected a little late.
- Cost: the CP is pure overhead**, up to 20 to 25 % of throughput.

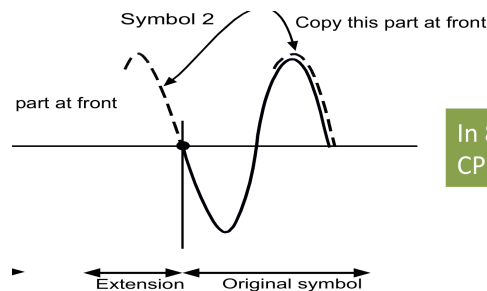
### Transmitter chain 4

- Modulation / symbol mapping:** bits  $\rightarrow$  QPSK or QAM symbols (after channel coding and interleaving).
- S/P:** serial symbol stream split into  $N$  parallel subcarrier values, i.e. a frequency-domain block.
- IFFT:** converts that block to  $N$  time-domain samples. This is the modulation.
- P/S:** back to a serial sample stream.
- Insert CP:** cyclic prefix prepended (see below).
- D/A**  $\rightarrow$  up-convert to RF  $\rightarrow$  antenna.

### Receiver chain 4

- Down-convert  $\rightarrow$  **A/D**.
- Synchronise:** packet detection, then carrier frequency offset correction.
- Remove CP**, discard the guard samples.
- S/P**  $\rightarrow$  block of  $N$  time samples.
- FFT**  $\rightarrow$  back to the frequency domain, one value per subcarrier.
- Equalize:** divide each subcarrier by its estimated  $H[k]$ , one tap each.
- P/S**  $\rightarrow$  **de-modulation / de-mapping**  $\rightarrow$  deinterleave and decode  $\rightarrow$  data out.

**Receiver is the transmitter read right to left.** Say that and you have drawn half the answer.



In 802.11,  
CP: data = 1:4

The tail of the symbol is copied to the front as an extension. AS Sir, OFDM notes

### Two more practical details

- Virtual (null) subcarriers** at the band edges are left unused: edge bins are the most vulnerable to frequency offset, and the gap suppresses out-of-band radiation.
- Pilot subcarriers** carry known symbols so the receiver can track residual phase and estimate  $H[k]$ .
- 802.11a splits its 64 bins as **48 data + 4 pilot + 12 null** verified/added. The lecture note's "48 of 64 occupied" counts only the data bins.

## OFDM: advantages and disadvantages 4

PIN 1/22 82 Bh 4+4 · asked together with “draw the transmitting side”, so budget the time

### Advantages

- **High spectral efficiency:** overlapping orthogonal subcarriers, no guard bands, roughly twice FDM's.
- **Immune to ISI and frequency-selective fading:** long symbols plus the CP.
- **Simple equalization:** one complex multiply per subcarrier, no adaptive transversal filter.
- **Efficient to build:** FFT/IFFT in DSP, cost grows as  $N \log N$ , not  $N^2$ .
- **Robust to narrowband interference:** a jammer kills a few subcarriers, and coding + interleaving recovers them.
- **Flexible:** power and modulation order can be set per subcarrier (adaptive bit loading); extends naturally to OFDMA and MIMO.

### Disadvantages

- **High PAPR (peak-to-average power ratio):**  $N$  subcarriers can add in phase, so the amplifier needs heavy back-off → poor efficiency. **The main drawback.** It is why LTE uses SC-FDMA, not OFDMA, on the uplink.
- **Very sensitive to frequency offset and phase noise:** any error destroys orthogonality → inter-carrier interference (ICI).
- **Sensitive to Doppler:** a fast-moving mobile changes the channel within one long symbol.
- **CP overhead** wastes 20 to 25 % of both time and energy.
- **Tight synchronisation** needed in both time and frequency.

Used in: DAB, DVB-T, 802.11a/g/n/ac, WiMAX, LTE downlink, 5G NR, ADSL.

## 4.7 Spread spectrum: PN sequences, DSSS and FHSS

### Define spread spectrum modulation; DSSS with transmitter and receiver block diagrams 5 6 2+6

HOT 4/22 82 Bh 2+6, 81 Ba 6, 80 Ch 2+5, 73 Ma 4 · spread spectrum overall: 7 of 22 papers

### Definition 2

- A modulation in which the transmitted signal occupies a bandwidth **far greater than the minimum needed to send the message.**
- Spreading is done by a **code independent of the data**, usually a PN sequence.
- The receiver **de-spreads** using a synchronised copy of the same code.
- **Only a receiver holding the code can recover the message.** To anyone else the signal looks like a rise in the noise floor.
- **Processing gain**  $PG = \frac{B_{ss}}{B} = \frac{T_b}{T_c} = N$  chips per bit. This number is the jamming margin.

### The two problems it answers

- How to secure information over a broadcast medium.
- How to survive deliberate jamming and external interference.
- (And, as a bonus, how to let many users share one band with no coordination.)

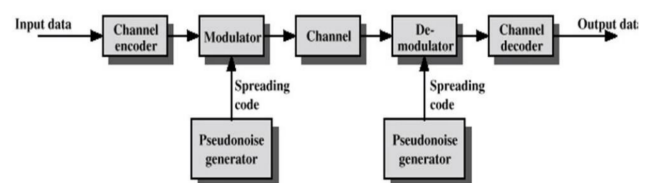


Figure 7.1 General Model of Spread Spectrum Digital Communication System

General model of a spread spectrum system. Note the symmetry: the same pseudonoise generator appears at both ends, and it sits outside the channel coding. SST Ch 4, Fig 7.1

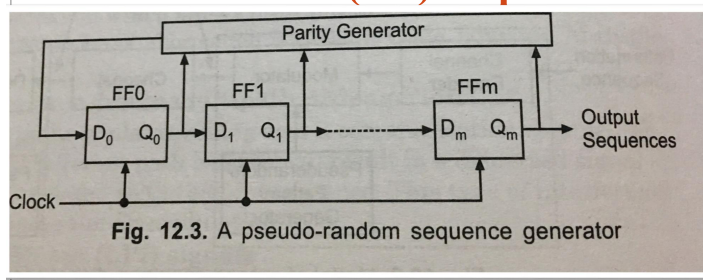


Fig. 12.3. A pseudo-random sequence generator

PN sequence generator:  $m$  D flip-flops in a shift register, XOR parity feedback into  $D_0$ . Maximal length  $2^m - 1$ . SST Ch 4, Fig 12.3

### DSSS transmitter 3

1. Original data bit stream at rate  $R_b$ , in  $\pm 1$  form.
2. **Chip (PN) generator** produces a code at rate  $R_c = NR_b$ , i.e.  $N$  chips per data bit.
3. **Multiplier** multiplies data by chips. **Each data bit is replaced by  $N$  chips**, which is where the bandwidth expansion happens.
4. The wideband product PSK-modulates the carrier  $\rightarrow$  transmitted DSSS signal.
5. Spectrum is now  $N$  times wider and  $N$  times lower in power density, so the signal can sit *beneath* the noise floor.

### DSSS receiver 3

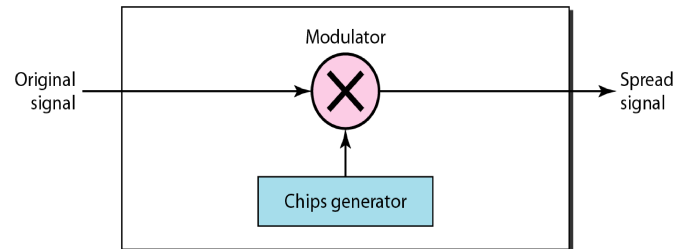
1. BPF and PSK demodulation give back the wideband chip stream.
2. A **local PN generator**, synchronised to the transmitter's, produces the same code.
3. **Multiply again by the same code**: since  $(\pm 1)^2 = +1$ , the data collapses back to its original narrow bandwidth (de-spreading).
4. **Anything not multiplied by the matching code gets spread instead of de-spread**, so interference and other users are pushed down by the processing gain.
5. Narrowband filter + integrate-and-dump  $\rightarrow$  recovered data.

### DSSS (+)

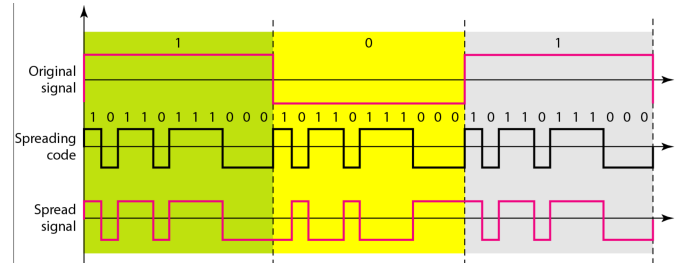
- Multiple access with no coordination  $\rightarrow$  **CDMA**, IS-95 and 3G.
- Narrowband jamming is suppressed by the full processing gain.
- Multipath components arriving more than one chip apart are resolvable  $\rightarrow$  **RAKE** receiver turns them into diversity (Ch 5).
- No frequency planning: every cell can use every frequency.

### PN sequence properties 2

- Coded sequence of 1s and 0s with **noise-like** statistics but deterministic generation.
- **Balance**: 1s and 0s occur with near-equal probability.
- **Shift property**: adding a shifted copy of a PN sequence gives back the same sequence in a different phase.
- **High autocorrelation** at zero lag, near zero elsewhere  $\rightarrow$  the receiver can find the exact code phase.
- **Low cross-correlation** with other codes  $\rightarrow$  other users look like noise.
- Easy to generate and to synchronise, in hardware, at chip rate.



DSSS spreading. SST Ch 4



DSSS waveform example: one data bit, an 11-chip code, and their product. Data 1 passes the code through, data 0 inverts it. SST Ch 4

### DSSS (-)

- **Near-far problem**: a close transmitter swamps a distant one, so tight and fast power control is mandatory.
- PN acquisition and tracking is slow and hardware-hungry.
- **Self-jamming**: code cross-correlation is low but not zero, so users interfere with each other as the load rises.

## FHSS operation with a neat block diagram 4 5

**HOT 3/22** 79 Ch 2+5, 73 Bh 4, 73 Ma 4 · as FHMA this is a TOP-tier Ch 7 question, see §7.3

### Operation 4

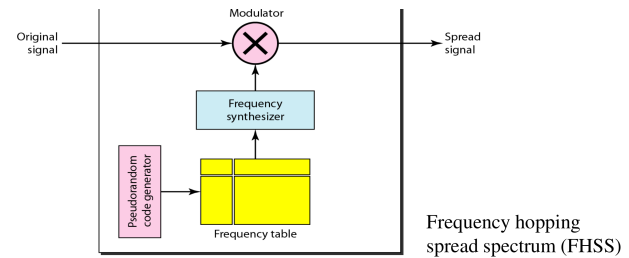
- FHSS uses  $M$  different carrier frequencies, and the carrier hops among them in a pseudo-random pattern.
- Binary data is applied to an **M-ary FSK modulator**.
- A **pseudorandom code generator** drives a **frequency synthesizer**, which produces the whole range of hop frequencies from one reference.
- The synthesizer output at any instant is the **frequency hop**, chosen by the current  $k$  code bits from a frequency table of  $2^k$  entries.
- A **mixer** combines each hop with the MFSK signal → transmitted signal.
- **The spectrum is spread sequentially, not instantaneously:** at any one moment the signal is narrowband, but over time it covers the whole band.
- The receiver runs the same code and synthesizer, so its local oscillator hops in lockstep and de-hops the signal back to a fixed IF.

### Slow against fast hopping

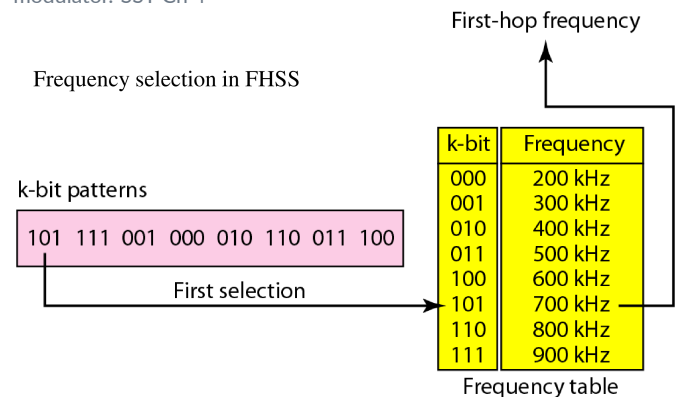
- **Slow FHSS:** several symbols per hop ( $R_h < R_s$ ). Cheaper synthesizer, used by GSM and Bluetooth.
- **Fast FHSS:** several hops per symbol ( $R_h > R_s$ ). Each symbol gets frequency diversity, so a bad hop cannot kill it. Costs a fast synthesizer.

### FHSS (+)

- **Immune to the near-far problem:** two users are almost never on the same frequency at the same instant, so no tight power control is needed.
- Resists partial-band jamming and frequency-selective fading, because the signal moves out of the bad band on the next hop.
- Synchronisation is easier and faster than DSSS's chip-level acquisition.
- Longer range for the same power, since instantaneous bandwidth stays narrow.



FHSS transmitter. Code generator → frequency table → synthesizer → modulator. SST Ch 4



Frequency selection: each 3-bit group of the PN stream indexes one row of the frequency table, which is the next hop. SST Ch 4

### FHSS (-)

- Needs a fast, agile and expensive **frequency synthesizer**.
- **Hits:** when two users pick the same hop, both symbols are lost, so error control coding is mandatory.
- Detection is usually non-coherent (the phase is lost at every hop), costing about 3 dB.
- Total bandwidth is large, so it must be licensed or confined to an ISM band.

Bluetooth: 79 hop channels, 1600 hops/s, 625  $\mu$ s dwell.

## Advantages and major disadvantages of spread spectrum 2 4

76 Bh 4 (+), 77 Ch 2 (-)

### Advantages 4

- **Anti-jam:** a narrowband jammer is spread by  $PG$  at the receiver, so its effective power drops by the processing gain.
- **Low probability of intercept:** the signal sits under the noise floor, so an eavesdropper cannot even tell it is there.
- **Privacy:** without the code the signal cannot be de-spread.
- **Multiple access:** near-orthogonal codes let many users occupy one band at once (CDMA).
- **Multipath rejection:** delayed copies decorrelate with the code and are either rejected or harvested by a RAKE receiver.
- **Graceful degradation:** adding users raises the noise floor a little rather than blocking calls outright.
- Mitigates ISI, and **bandwidth efficiency actually rises as simultaneous users increase.**

### Major disadvantages 2

- **Large bandwidth** needed, many times the message bandwidth. Spectrum is expensive.
- Complex and costly transmitter and receiver: PN generation, acquisition, tracking.
- **Long acquisition time** before the first bit can be decoded.
- **Near-far problem** in DSSS → fast closed-loop power control needed.
- **Self-jamming** from non-zero code cross-correlation, which caps capacity.
- Synchronisation of the two PN generators must be exact to within a fraction of a chip.

## 4.8 Performance of modulation in fading channels syllabus, no PYQ yet

### What the fading channel does to BER

- In **AWGN** the error probability falls *exponentially* with  $E_b/N_0$ .
- In **flat Rayleigh fading** it falls only *inversely*:  $\bar{P}_e \approx 1/(4 E_b/N_0)$  for coherent BPSK.
- Consequence: reaching  $10^{-3}$  BER needs about **7 dB in AWGN and about 24 dB in Rayleigh fading**. Tens of dB of extra power for the same quality. verified/added
- Cause: the deep fades dominate the average, not the typical signal level.
- Cure is **diversity**, not more power (Ch 5).

### Which scheme survives the mobile channel

- **Coherent** schemes suffer most: the receiver cannot track a phase reference that is being randomised by the channel.
- **Differential and non-coherent** schemes (DPSK,  $\pi/4$ -DQPSK, GMSK with a discriminator) degrade far more gracefully.
- **Constant envelope** schemes suit the handset, because the power amplifier can saturate.
- **Linear** schemes (QAM) need back-off and accurate gain control, so they belong on the downlink or on a fixed link.

### The irreducible error floor

- In **frequency-selective** fading, ISI produces errors that **no amount of transmit power removes**.
- The floor is set by the ratio  $\sigma_\tau/T_s$ , i.e. delay spread against symbol period.
- In **fast fading**, random FM from Doppler produces a second floor set by  $f_d T_s$ .
- Four ways out: equalization, spread spectrum + RAKE, OFDM (long symbols), and diversity.

### The trade-off table in one line each

- **BPSK:** most power efficient binary scheme, poorest  $\eta_B$ .
- **QPSK / OQPSK:** double  $\eta_B$  at no BER cost, envelope problems.
- **MSK / GMSK:** constant envelope, non-coherent detection, GSM's choice.
- **M-QAM:** highest  $\eta_B$ , needs high SNR and a linear amplifier.
- **DSSS:** trades a huge bandwidth for interference immunity and multiple access.
- **OFDM:** turns a selective channel into many flat ones, pays in PAPR.

## 4.9 Last-minute recall for this chapter

### Must memorise

- QPSK: 2 bits/symbol,  $T_s = 2T_b$ , null-null BW =  $R_b$ , **same BER as BPSK**.
- MSK: CPFSK with  $h = 0.5$ ,  $\Delta f = R_b/4$ , equals OQPSK with half-sine pulses.
- GMSK: MSK + Gaussian LPF,  $H(f) = \exp[-\frac{\ln 2}{2}(f/B)^2]$ , **GSM uses  $BT = 0.3$** .
- Smaller  $BT \rightarrow$  tighter spectrum, more ISI.
- OFDM:  $\Delta f = 1/T$ , IFFT at the transmitter, FFT at the receiver, CP > delay spread, main drawback **PAPR**.
- Spread spectrum:  $PG = B_{ss}/B = T_b/T_c = N$ .
- PN generator:  $m$  stages, maximal length  $2^m - 1$ .
- 16-QAM against 16-PSK: **1.6 dB less peak power** at equal  $d_{\min}$ , 4.19 dB less average power.
- $\pi/4$ -QPSK: max phase change  $135^\circ$ , differentially detectable, IS-136 and PDC.

**No paper in 2070–2082 has set a numerical from this chapter.** The one calculation worth holding is the Gaussian-filter sizing in §4.4, which is a tutorial problem, not a PYQ. Everything examined here is descriptive, and the marks live in the figures.

### Abbreviations used in these notes

Only these are ever shortened. Anything not on this list is written out in full.

### Most repeated, in order

1. OFDM transmitter and receiver +Fig **TOP 9/22**
2. Spread spectrum, DSSS and FHSS +Fig **TOP 7/22**
3. MSK and GMSK, and why GSM chose GMSK **TOP 6/22**
4. QPSK transmission and detection +Fig **HOT 4/22**
5. OFDM (+) and (-) **PIN 1/22**
6. 16-QAM against 16-PSK **PIN 1/22**
7. DPSK transmitter and receiver, PN sequence **PIN 1/22**
8. OQPSK, and why  $\pi/4$ -QPSK beats it **PIN 1/22**

**Draw first, then annotate.** An unlabelled correct block diagram still scores; three paragraphs with no diagram do not.

| Short    | Means                                       | Short    | Means                                      |
|----------|---------------------------------------------|----------|--------------------------------------------|
| BS       | base station                                | BTS      | base transceiver station                   |
| MS       | mobile station                              | MSC      | mobile switching centre                    |
| BSC      | base station controller                     | MSK      | minimum shift keying                       |
| CCI      | co-channel interference                     | ACI      | adjacent channel interference              |
| SIR, S/I | signal to interference ratio                | SNR, S/N | signal to noise ratio                      |
| GoS      | grade of service                            | MAHO     | mobile assisted handoff                    |
| FCA      | fixed channel assignment                    | DCA      | dynamic channel assignment                 |
| LOS      | line of sight                               | NLOS     | non line of sight                          |
| FDMA     | frequency division multiple access          | TDMA     | time division multiple access              |
| CDMA     | code division multiple access               | OFDM     | orthogonal frequency division multiplexing |
| FDD      | frequency division duplex                   | TDD      | time division duplex                       |
| GSM      | global system for mobile communications     | AMPS     | advanced mobile phone system               |
| PCS      | personal communication system               | DAS      | distributed antenna system                 |
| BCCH     | broadcast control channel                   | EIRP     | effective isotropic radiated power         |
| ERP      | effective radiated power                    | RF       | radio frequency                            |
| Tx, TX   | transmitter                                 | Rx, RX   | receiver                                   |
| ISI      | intersymbol interference                    | PDP      | power delay profile                        |
| DFE      | decision feedback equalizer                 | MLSE     | maximum likelihood sequence estimation     |
| LMS      | least mean square                           | RLS      | recursive least squares                    |
| MSE      | mean square error                           | MMSE     | minimum mean square error                  |
| ZF       | zero forcing                                | MRC      | maximal ratio combining                    |
| EGC      | equal gain combining                        | BER      | bit error rate                             |
| PCM      | pulse code modulation                       | DPCM     | differential PCM                           |
| ADPCM    | adaptive differential PCM                   | LPC      | linear predictive coder                    |
| RPE-LTP  | regular pulse excited, long term prediction | CELP     | code excited linear prediction             |
| RELTP    | residual excited linear prediction          | MPE      | multipulse excited                         |
| VSELP    | vector sum excited linear prediction        | ATC      | adaptive transform coding                  |
| SBC      | sub-band coding                             | DCT      | discrete cosine transform                  |
| FEC      | forward error correction                    | ARQ      | automatic repeat request                   |
| BCH      | Bose–Chaudhuri–Hocquenghem code             | RS       | Reed–Solomon code                          |
| TCM      | trellis coded modulation                    | PN       | pseudo-noise                               |

| Short      | Means                                                  | Short         | Means                                               |
|------------|--------------------------------------------------------|---------------|-----------------------------------------------------|
| MIMO       | multiple input multiple output                         | OFDMA         | orthogonal FDM access                               |
| QAM        | quadrature amplitude modulation                        | PSK           | phase shift keying                                  |
| GMSK       | Gaussian minimum shift keying                          | IMT           | international mobile telecommunications             |
| ASK        | amplitude shift keying                                 | FSK           | frequency shift keying                              |
| BPSK       | binary phase shift keying                              | QPSK          | quadrature phase shift keying                       |
| OQPSK      | offset (staggered) QPSK                                | DPSK,         | differential (binary) PSK                           |
|            |                                                        | DBPSK         |                                                     |
| CPFSK      | continuous phase FSK                                   | MFSK,         | M-ary FSK, M-ary PSK                                |
|            |                                                        | MPSK          |                                                     |
| DSSS       | direct sequence spread spectrum                        | FHSS          | frequency hopped spread spectrum                    |
| SS, SSM    | spread spectrum (modulation)                           | PG            | processing gain                                     |
| PAPR       | peak-to-average power ratio                            | CP            | cyclic prefix                                       |
| FFT, IFFT  | (inverse) fast Fourier transform                       | ICI           | inter-carrier interference                          |
| PSD        | power spectral density                                 | CFO, SFO      | carrier, sampling frequency offset                  |
| NRZ, RZ    | non-return, return to zero                             | AMI           | alternate mark inversion                            |
| AM, FM, PM | amplitude, frequency, phase modulation                 | DSB, SSB, VSB | double, single, vestigial sideband                  |
| S/P, P/S   | serial-parallel converter                              | D/A, A/D      | digital-analog converter                            |
| LPF, BPF   | low pass, band pass filter                             | ISM           | industrial, scientific, medical band                |
| SC-FDMA    | single carrier FDMA                                    | <i>BT</i>     | GMSK filter bandwidth × symbol time                 |
| PSTN       | public switched telephone network                      | VoIP          | voice over IP                                       |
| FVC, RVC   | forward, reverse voice channel                         | FCC, RCC      | forward, reverse control channel                    |
| SSMA       | spread spectrum multiple access                        | SDMA          | space division multiple access                      |
| FHMA       | frequency hopped multiple access                       | DSMA          | direct sequence multiple access (= CDMA)            |
| FCDMA      | hybrid FDMA/CDMA                                       | DS/FHMA       | hybrid direct sequence / frequency hopped           |
| TCDMA      | time division CDMA                                     | TDFH          | time division frequency hopping                     |
| IM         | intermodulation (product)                              | MU-MIMO       | multi-user MIMO                                     |
| WLAN       | wireless local area network                            | WMAN          | wireless metropolitan area network                  |
| WPAN       | wireless personal area network                         | AP            | access point                                        |
| BSS, IBSS  | basic service set, independent (ad hoc) BSS            | ESS           | extended service set                                |
| CSMA/CA    | carrier sense multiple access with collision avoidance | CSMA/CD       | ... with collision detection                        |
| DCF        | distributed coordination function                      | PCF           | point coordination function                         |
| RTS/CTS    | request to send / clear to send                        | CCK           | complementary code keying                           |
| WEP, WPA   | wired equivalent privacy, WiFi protected access        | U-NII         | unlicensed national information infrastructure band |
| IoT        | internet of things                                     | mmWave        | millimetre wave                                     |
| eMBB       | enhanced mobile broadband                              | URLLC         | ultra-reliable low latency comm.                    |
| mMTC       | massive machine type comm.                             | NOMA          | non-orthogonal multiple access                      |
| D2D        | device to device                                       | C-RAN         | cloud radio access network                          |
| PCN        | personal communication network                         | SMS           | short message service                               |
| attr(s)    | attribute(s)                                           | no.           | number                                              |
| freq       | frequency                                              | info          | information                                         |
| ops        | operations                                             | config        | configuration                                       |
| rm         | remove                                                 | vs, Vs        | versus                                              |
| w/, w/o    | with, without                                          | Eg            | example / for example                               |
| +Fig       | draw a figure                                          | (I)           | importance / need / significance                    |
| →          | leads to / therefore                                   | +             | and                                                 |

Standard and system names (proper names, kept as printed): **AMPS** advanced mobile phone system · **ETACS** extended total access communication system · **NMT** Nordic mobile telephone · **NTT** Nippon Telegraph and Telephone · **IMTS** improved mobile telephone service · **PDC** Pacific digital cellular · **IS-95** cdmaOne · **IS-136** digital AMPS · **HSCSD** high speed circuit switched data · **GPRS** general packet radio service · **EDGE** enhanced data rates for GSM evolution · **W-CDMA** wideband CDMA · **UMTS** universal mobile telecommunications system · **cdma2000 1xEV-DO** / **1xEV-DV** the cdma2000 3G family · **LTE**, **LTE-A** long term evolution (advanced) · **WiMAX** IEEE 802.16 · **WiFi** IEEE 802.11 · **DECT** digital enhanced cordless telecommunications · **CT2** cordless telephone 2 · **PHS** personal handyphone system · **USDC** US digital cellular (IS-136) · **JDC** Japanese digital cellular (later PDC) · **Bluetooth** IEEE 802.15.1 · **Zigbee** IEEE 802.15.4 · **PACS** personal access communications system · **POCSAG**, **FLEX** paging protocols · **CDPD** cellular digital packet

data · **Mobitex** a dedicated packet data system · **TETRA** terrestrial trunked radio · **DAB** digital audio broadcasting · **DVB-T** digital video broadcasting, terrestrial · **ADSL** asymmetric digital subscriber line · **5G NR** 5G new radio.

**One clash to watch:** **FCC** is the *forward control channel* throughout these notes. The US regulator is always written out as the Federal Communications Commission.

**Exam-paper month codes** (used in the year tags): Ba = Baishakh · Bh = Bhadra · Ash = Ashwin · Ma = Magh · Ch = Chaitra.

**Bold** year tags are Regular exams, plain are Back exams. Monospace year tags are BEI (EX 715); the rest are BEX (EX 751).

**Tier chips** over 22 papers, 2070–2082 BS: **TOP /22** = topic asked in 6 or more papers · **HOT /22** = 3 to 5 papers · **PIN /22** = 1 or 2 papers but worth  $\geq 6$  marks. **Mark chips** such as **2+5** are the marks that question actually carried. **syllabus, no PYQ yet** marks a syllabus topic that has never been examined but is written up anyway. **verified/added** marks anything researched and added that was not in the source notes.